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MagZiCw66iU6-w= (Other)

Top Skills

Basic Life Support (BLS)
Automated External Defibrillator
(AED)
Cardiopulmonary Resuscitation
(CPR)

Languages

Kannada (Full Professional)
English (Native or Bilingual)
Telugu (Native or Bilingual)
Hindi (Elementary)

Certifications

Conflicts of Interest
Google Professional Workspace
Administrator Specialization
Foundational C# with Microsoft
JavaScript Algorithms and Data
Structures
Basic Life Support (BLS) Provider

Honors-Awards

Florence Leas Sophomore Prize
First Place - 32nd Annual Spring
Programming Contest
Spring 2022 Dean's List
2022-2023 WCSA Best New
Member - Golden Bishop Award
Spring 2023 Dean's List

Publications

Aadarsha Gopala Reddy

Seeking Data/Machine Learning Engineer Roles | WashU MS CS '26
St Louis, Missouri, United States

Summary

I build and evaluate machine learning systems where the data doesn't come clean. My graduate research at Washington University in St. Louis asked a question at the intersection of AI and neuroscience: can everyday driving behavior reveal early signs of cognitive decline? To answer it, I built an end-to-end pipeline processing 26,968 participant-weeks of naturalistic driving telemetry from 304 older adults across 428 unique vehicles and compared six model families, from logistic regression to domain-adversarial GRUs, under leave-one-participant-out evaluation. The signal was subtle, but the process taught me how to build and rigorously validate ML pipelines under real-world constraints. My industry experience began with internships that covered the full ML lifecycle. As an AI Engineer at Crittero, Inc., I engineered a full-stack social media simulation and recommendation system, building persona-based data generation in TypeScript and a recommendation algorithm using TensorFlow. At Lab714, I built React and AWS dashboards for IoT data that helped drive a 20% cost reduction for clients. On the side, I've implemented deep reinforcement learning (DDQN, PPO, SAC) for datacenter cooling optimization, achieving a 35.8% energy efficiency improvement over baseline controllers. I've also spent significant time in roles that taught me to communicate technical work across audiences. As a Graduate Teaching Assistant for WashU's Data Manipulation and Management at Scale course, I guided students through building data pipelines with Python, Snowflake, SQL, Airflow, Spark, Kafka, and Flink, and conducted oral exams modeled after technical interviews. At the Taylor Family Center for Student Success, I led data storytelling and analytics initiatives that helped university staff make evidence-based decisions. I served as Vice President of the Graduate-Professional Council (GPC) Chamber of the Graduate and Professional Student Council at WashU, chaired its Constitution Committee, and sat on two university advisory boards — experiences that gave me a firsthand look at how institutions make decisions and how to advocate effectively. I recently completed my MS in Computer Science at WashU (May 2026) and am looking

for software engineering, data engineering, or machine learning roles. If your team works on technically interesting problems, I'd welcome a message here or at adurs2002@gmail.com

Experience

Washington University McKelvey School of Engineering

1 year 4 months

Graduate Teaching Assistant - CSE 5114: Data Manipulation and Management at Scale

January 2026 - May 2026 (5 months)

St Louis, MO

- Facilitated a graduate-level course focused on the methodologies and technologies essential for managing and manipulating substantial data volumes efficiently and responsibly.
- Guided students in the implementation of data pipelines and real-time processing systems using industry-standard technologies, including Python, SQL, Apache Airflow, Spark, Kafka, Flink, and Snowflake.
- Conducted 20-25 minute oral midterm exams designed to simulate technical data systems design interviews, assessing student mastery of distributed storage, fault tolerance, and scaling.
- Mentored 2-4 student groups through semester-long capstone projects, providing architectural advice on collecting, cleaning, and interpreting data from non-trivial real-world sources.
- Reinforced key engineering concepts such as data warehousing, NoSQL databases, denormalization techniques, and data privacy principles like de-identification.
- Managed asynchronous technical support via Ed and held bi-weekly office hours to assist students with debugging code and configuring distributed systems tools.

McKelvey Engineering Graduate Ambassador

February 2025 - May 2026 (1 year 4 months)

St Louis, MO

- Represented McKelvey Engineering at conferences and events to share WashU experiences with prospective graduate students.
- Developed leadership skills by serving as an official student representative to support new and current students.
- Utilized email, virtual events, and in-person campus events to engage with students and provide support.

- Collaborated with teams to enhance the graduate student experience and promote McKelvey Engineering's programs.

Washington University in St. Louis

Graduate Assistant with the Taylor Family Center for Student Success

November 2025 - May 2026 (7 months)

St Louis, MO

- Led data analysis and storytelling initiatives to measure program impact by analyzing attendance, student demographics, and pre/post-program surveys.
- Administered the TFCSS website, implementing new technical functionalities such as submission forms and staff scheduling tools, while managing content updates and performing site audits.
- Improved long-term administrative systems by developing detailed process manuals for web and social media and training TFCSS staff on data evaluation best practices.
- Enhanced marketing and outreach strategy by analyzing social media and newsletter engagement data, creating "Spotlight Stories", and partnering with SA Marcomm on web analytics.

Crittero, Inc.

AI Engineer Intern

June 2025 - August 2025 (3 months)

- Built a robust social media simulation system generating realistic training data with 7 distinct persona types, seeded random generation, and comprehensive activity modeling across 1700+ lines of TypeScript code.
- Developed foundational recommendation algorithm framework with neural network architecture and collaborative filtering fallback, learning TensorFlow/Keras fundamentals while identifying areas needing reliability improvements.
- Created extensive testing infrastructure with performance metrics, reproducibility validation, and configurable demo modes across 700+ lines of Python test code, establishing debugging practices for complex ML systems.
- Implemented a comprehensive data ingestion pipeline successfully processing CSV exports, and interaction matrices with proper data flow validation.
- Designed persona-based behavioral modeling with time-series patterns, weighted probabilities, and activity logging levels, achieving sophisticated user simulation capabilities.
- Learned TypeScript/Node.js ecosystem challenges while building complex object-oriented systems, gaining appreciation for Python's simplicity and discovering personal technology preferences.

- Established Git workflow practices and collaborative development processes while contributing to a multi-developer codebase with proper version control.
- Debugged interaction matrix creation issues and identified reliability gaps in recommendation scoring, developing critical problem-solving skills for production ML systems.
- Built a configurable CLI testing framework with customizable parameters enabling systematic evaluation of algorithm performance and reproducible experiment execution.
- Gained hands-on experience with machine learning model training, data preprocessing, synthetic data generation, and the complexities of building recommendation systems from scratch.

Washington University in St. Louis

Graduate Assistant with the Taylor Family Center for Student Success
August 2024 - May 2025 (10 months)

St Louis, Missouri, United States

- Led and coordinated strategic initiatives designed to foster success for First-Generation, Limited-Income (FLI) students, including the planning and execution of key community-building programs like "End of Week Unwind."
- Drove marketing and communication efforts for the Taylor Family Center by managing social media channels and co-developing targeted newsletters for alumni and the Forward Families program.
- Pioneered data assessment efforts by tracking and analyzing user interaction on Instagram and newsletters; translated data-driven insights into actionable strategies that led to measurable improvements in engagement.
- Served as a trusted representative for the department on behalf of the Executive Director and provided integral operational support to the Taylor Family Center leadership team.

Lab714

Data Analyst

June 2023 - May 2024 (1 year)

Boca Raton, Florida, United States

- Achieved a 20% reduction in operational costs for clients by identifying key inefficiencies and actionable insights through in-depth analysis of IoT sensor data.
- Engineered and deployed an end-to-end IoT data management system on AWS, featuring a robust data pipeline (leveraging S3, Lambda) for automated real-time data collection and processing, coupled with a React-

based application for streamlined data analysis and visualization, ensuring data integrity and accessibility.

- Collaborated with a cross-functional team to gather requirements, design solutions, and implement data analysis strategies, demonstrating effective teamwork and communication skills.

Ohio Wesleyan University

1 year 9 months

Computer Science Lab Assistant

January 2022 - May 2023 (1 year 5 months)

Delaware, Ohio, United States

- Tutored around 10 students for over 6 hours per week, aiding with homework and exam preparation for intro, intermediate and some high-level coursework.
- Tested and graded around 35 students' homework for introductory and intermediate computer science classes.

Media Center Assistant

September 2021 - May 2023 (1 year 9 months)

Delaware, Ohio, United States

- Setup and tore down audio-visual equipment, logged equipment movement and created appointments for use of spaces for approximately 9 hours per week.
- Assisted with providing AV systems and support for events while ensuring successful setup and execution.

Computer Science Researcher

May 2022 - July 2022 (3 months)

Delaware, Ohio, United States

- Developed and tested a digital version of the card game Lost Cities using JAVA in 2 weeks.
- Implemented an intelligent agent that played the game against a human, winning 13 of 18 games in testing.
- Presented at the Patricia Belt Conrades Summer Science Research Symposium as part of OWU Connection Conference.

Reunion Weekend Ambassador

May 2022 - May 2022 (1 month)

Delaware, Ohio, United States

- Organized events focused toward Ohio Wesleyan Alumni.

- Assisted the Alumni Relations department coordinate, prepare, and conduct Reunion Weekend from May 13 to May 15 with an attendance of over 600 alumni.
- Oversaw Class of 1973's 50th Reunion events, interacted with alumni and learned about their time as a student.

Peer Tutor

September 2021 - January 2022 (5 months)

Delaware, Ohio, United States

- Tutored and provided resources to 4 "Introduction to Environment and Sustainability" course students to keep up with assignments and course content for over 4 hours every week.

The Next Genius Foundation

Interviewer

November 2021 - December 2021 (2 months)

Supported the scholarship foundation in interviewing and mentoring 10 applicants for 2022 undergraduate university scholarships.

The Next Genius Foundation

Interviewer

December 2020 - January 2021 (2 months)

- Supported the scholarship foundation in interviewing and mentoring 10 applicants for 2021 undergraduate university scholarships.
- Assisted with the data entry of student information into a database.

Education

Washington University in St. Louis

Master of Science - MS, Computer Science · (May 2026)

Ohio Wesleyan University

Bachelor of Arts - BA, Computer Science & Data Analytics; Economics minor · (August 2020 - May 2023)

Indus International School Bangalore

International Baccalaureate Diploma Programme · (May 2018 - May 2020)

Ebenezer International School Bangalore

Indian Certificate of Secondary Education · (May 2018)